Kangsheng Qi

qikangsheng@gmail.com | +41 76 449 75 30 | Zurich, Switzerland | Nationality: Chinese

Research Interests

- Computational modeling and analysis of biological networks
- Interpretability and explainability of biomedical models

Education

ETH Zurich, Zurich, Switzerland 09/2025 – Present
M.Sc. in Computer Science

University of Texas at Austin, Austin, USA 08/2022 – 08/2025
B.Sc. in Mathematics (Honors)
B.Sc. in Computer Science (Departmental Honors)
Advisor: Prof. Can Cenik

Thesis

[1] **Kangsheng Qi**. Graph-based Approaches for Protein Network Analysis and Function Prediction. *Bachelor's Thesis*. The University of Texas at Austin. 2025. Link: https://apps.cs.utexas.edu/apps/tech-reports/206110

Research Experience

Martin Zhang Lab, Carnegie Mellon University 06/2026 – Present
Visiting Graduate Researcher | Advisor: Prof. Martin Jinye Zhang

- Integrate cell type information into KGWAS, a knowledge graph-based machine learning model to enhance genome-wide association study
- Improve KGWAS stability and develop robust methods for interpreting disease-relevant links and uncovering underlying biological mechanisms

Biomedical Informatics Group, ETH Zurich 09/2025 – Present
Graduate Researcher | Advisor: Prof. Gunnar Rätsch

- Train sparse autoencoders (SAEs) to interpret clinical time-series foundation models
- Design an intervention framework to steer SAE features to control model predictions

Can Cenik Lab, University of Texas at Austin 01/2024 – 09/2025
Undergraduate Researcher | Advisor: Prof. Can Cenik

- Contributed analyses of large-scale transcriptomic and translational datasets to the study introducing translation efficiency covariation (TEC)
- Characterized the topology and functional properties of a human TEC network

The Physiological Laboratories, Dell Medical School 06/2024 – 08/2024
Summer Research Intern | Advisor: Prof. Joshua Chang

- Evaluated deep learning models for heart rate time-series forecasting in preterm infants

Publications

[2] **Kangsheng Qi**, Can Cenik. (2025). Topological and Functional Characterization of Human Translation Efficiency Covariation Network. *Bioinformatics*. doi: 10.1093/bioinformatics/btaf583.

[1] Yue Liu, Shilpa Rao, Ian Hoskins, Michael Geng, Qiuxia Zhao, Jonathan Chacko, Vighnesh Ghatpande, **Kangsheng Qi**, Logan Persyn, Jun Wang, Dinghai Zheng, Yochen Zhong, Dayea Park, Elif Sarinay Cenik, Vikram Agarwal, Hakan Ozadam, Can Cenik. (2025). Translation efficiency covariation identifies conserved coordination patterns across cell types. *Nature Biotechnology*. doi: 10.1038/s41587-025-02718-5.

Manuscripts

[1] **Kangsheng Qi**, Manuel Burger, Moritz Miller, Gunnar Rätsch. (2026). Interpreting and Steering Clinical Foundation Models through Sparse Autoencoders. *Under review at IEEE Conference on Secure and Trustworthy Machine Learning.*

Presentations

[2] **Kangsheng Qi**, Can Cenik. (2025). Graph-based Approaches for Protein Network Analysis and Function Prediction. Presented by K.Q. as poster at *MBS Retreat.*

[1] **Kangsheng Qi**, Joshua Chang. (2024). Deep Learning Approaches for Time Series Forecasting of NICU Heart Rate Data. Presented by K.Q. as poster and talk at *TACCSTER Symposium.* doi: 10.26153/tsw/55885.

Awards and Honors

Undergraduate Research Fellowship | $1,000 12/2024
Awarded to support research on modeling physiological signals in preterm infants.

Moncrief Summer Research Fellowship | $8,000 06/2024
Supported undergraduate research project in the Physiological Laboratories at Dell Medical School, supervised by Prof. Joshua Chang.

College Scholar 2024, 2025
Ranked in the top 20% of class by GPA within the College of Natural Science.

Teaching

MAS ETH in AI and Digital Technology, ETH Zurich 09/2026 – Present
Teaching Assistant | Instructor: Dr. Carlos Cotrini

Probability I, University of Texas at Austin 08/2023 – 12/2023
Course Assistant | Instructor: Prof. John Luecke

Languages

English: Fluent (C2 / TOEFL iBT 119)
Chinese: Native